- A higher code density than other 32-bit RISC processors in the market

SPARC

SPARC or Scalable Processor Architecture was a RISC based Instruction Set Architecture, developed by Sun Microsystems in the year 1987.

A separate company dubbed the SPARC International was established in 1989 to promote the use of the architecture and manage the SPARC trademarks and licenses.

It was licensed to a number of manufacturers including Texas Instruments and Fujitsu. Currently its design is fully open source and anyone can use it without having to pay any royalty.

The 'Scalable' part in SPARC is there because its architecture allows one to implement it on any scale, from embedded processors to large mainframe/ server processors, with each based on the same core instruction set.

A number of SoCs based on the architecture have been manufactured by a number of companies, powering up workstations and servers. Currently two companies, Fujitsu and Oracle, are developing SoCs based on the SPARC architecture.

Oracle, the current owner of Sun Microsystems, designed M7 from ground up. It was first announced in 2015 to power high end servers. The systems are powered by 32 core, 256 thread SPARC microprocessors, and have many new advancements which include the following features:

ORACLE®

Oracle bought Sun Microsystems and built M7, a new SPARC based processor from the ground up.

- Enhanced and secure memory protection technology called the "silicon secured memory", to stop malicious programs from accessing parts of the memory they shouldn't
- Hardware assisted encryption built into all the 32 cores
- Coprocessor capable of offloading functions and tasks from main core to speeds up a variety of tasks and critical functions like decompression, memory scan and range scan.